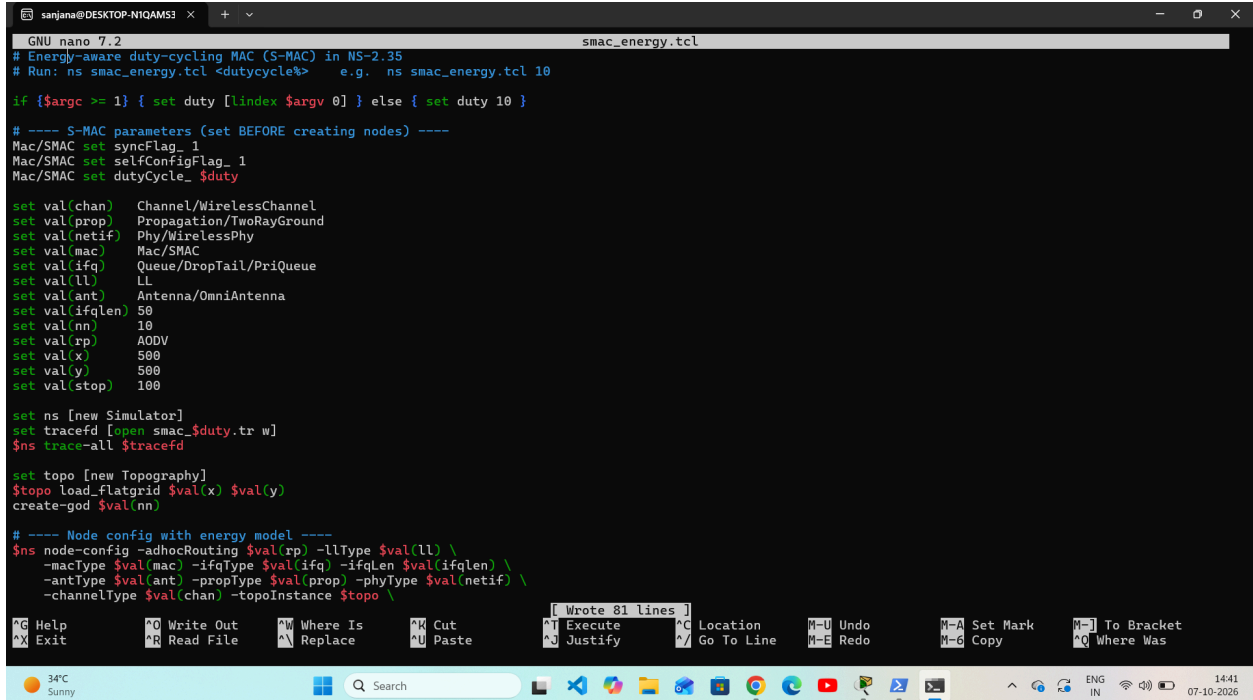


PRACTICAL NO: 18

AIM:Energy-Aware MAC Protocol for WSN
Implement duty-cycling MAC protocol and evaluate energy conservation in sensor networks.

```
sanjana@DESKTOP-N1QAMS3:~$ ns
% exit
sanjana@DESKTOP-N1QAMS3:~$ echo 'puts [Mac/SMAC info class]; exit' > t.tcl
sanjana@DESKTOP-N1QAMS3:~$ ns t.tcl
Class
sanjana@DESKTOP-N1QAMS3:~$ mkdir ~/wsn_practical
sanjana@DESKTOP-N1QAMS3:~$ cd ~/wsn_practical
sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ cp /mnt/c/Users/<username>/Downloads/smac_energy.tcl
```

nano smac_energy.tcl



```
GNU nano 7.2 smac_energy.tcl
# Energy-aware duty-cycling MAC (S-MAC) in NS-2.35
# Run: ns smac_energy.tcl <dutycycle%> e.g. ns smac_energy.tcl 10

if {$argc >= 1} { set duty [lindex $argv 0] } else { set duty 10 }

# ---- S-MAC parameters (set BEFORE creating nodes) ----
Mac/SMAC set syncFlag_ 1
Mac/SMAC set selfConfigFlag_ 1
Mac/SMAC set dutyCycle_ $duty

set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/SMAC
set val(ifq) Queue/DropTail/PriQueue
set val(ll) LL
set val(ant) Antenna/OmniAntenna
set val(ifqlen) 50
set val(nn) 10
set val(rp) AODV
set val(x) 500
set val(y) 500
set val(stop) 100

set ns [new Simulator]
set tracefd [open smac_$duty.tr w]
$ns trace-all $tracefd

set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)
create-god $val(nn)

# ---- Node config with energy model ----
$ns node-config -adhocRouting $val(rp) -llType $val(ll) \
-macType $val(mac) -ifqType $val(ifq) -ifqLen $val(ifqlen) \
-antType $val(ant) -propType $val(prop) -phyType $val(netif) \
-channelType $val(chan) -topoInstance $topo \
```

```
sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ nano smac_energy.tcl
sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ ls
smac_energy.tcl
sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ ns smac_energy.tcl 100
num_nodes is set 10
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
Starting simulation, duty cycle = 100 %

NOTE: SMAC is running with sleep-wakeup cycles on. Please make sure to run yr applications AFTER the nodes get sync'ed which is about 40sec for the default settings.

channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
Segmentation fault (core dumped)
```

```
sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ ns smac_energy.tcl 10
num_nodes is set 10
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
Starting simulation, duty cycle = 10 %

NOTE: SMAC is running with sleep-wakeup cycles on. Please make sure to run yr applications A
settings.

channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
Segmentation fault (core dumped)
```

PRACTICAL NO: 18

```
sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ sed -i 's/^set val(rp).*/set val(rp)
> DSDV/' smac_energy
sed: -e expression #1, char 28: unterminated 's' command
sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ sed -i 's/^set val(rp).*/set val(rp)      DSDV/' smac_energy.tcl
sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ sed -i 's/-macTrace ON/-macTrace OFF/' smac_energy.tcl
sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ grep -n "val(rp)\\|macTrace" smac_energy.tcl
20:set val(rp)      DSDV
34:$ns node-config -adhocRouting $val(rp) -llType $val(ll) \
38:      -agentTrace ON -routerTrace ON -macTrace OFF -movementTrace OFF \

sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ ns smac_energy.tcl 100
num_nodes is set 10
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
Starting simulation, duty cycle = 100 %

NOTE: SMAC is running with sleep-wakeup cycles on. Please make sure to run yr appl
settings.

channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
Segmentation fault (core dumped)

sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ sed -i 's/^set val(rp).*/set val(rp)      AODV/' smac_energy.tcl
sanjana@DESKTOP-N1QAMS3:~/wsn_practical$ grep -n "val(mac)\\|val(rp)" smac_energy.tcl
14:set val(mac)      Mac/SMAC
20:set val(rp)      AODV
34:$ns node-config -adhocRouting $val(rp) -llType $val(ll) \
35:      -macType $val(mac) -ifqType $val(ifq) -ifqLen $val(ifqlen) \
```

set ns [new Simulator]

Trace Files

set tr [open energy.tr w]

\$ns trace-all \$tr

set nf [open energy.nam w]

\$ns namtrace-all-wireless \$nf 500 500

Topography

set topo [new Topography]

\$topo load_flatgrid 500 500

create-god 5

Channel

set chan [new Channel/WirelessChannel]

Node Configuration

\$ns node-config \

-adhocRouting DSDV \

-llType LL \

-macType Mac/802_11 \

-ifqType Queue/DropTail/PriQueue \

-ifqLen 50 \

-antType Antenna/OmniAntenna \

-propType Propagation/TwoRayGround \

-phyType Phy/WirelessPhy \

-channel \$chan \

-topoInstance \$topo \

PRACTICAL NO: 18

